

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Fourth Semester B. Tech (CBCS) Degree Examinations

Summer Semester July 2024

EC2101-1 - COMMUNICATION SYSTEMS - I

(EC)

Duration: 3 Hours

Max. Marks:100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- For a message signal $m(t) = \cos(2\pi f_m t)$ and carrier signal $c(t) = \cos(2\pi f_c t)$, which of the following represents a single sideband (SSB) signal ?
 A) $\cos[2\pi(f_c + f_m)t]$ B) $\cos(2\pi f_c t)$
 C) $\cos(2\pi f_m t) \cos(2\pi f_c t)$ D) $[1 + \cos(2\pi f_m t)]\cos(2\pi f_c t)$
- The frequency discriminator consists of a slope circuit followed by _____.
 A) High pass filter B) Envelope detector
 C) BPF D) Non linear device
- Consider an angle modulated signal $x(t) = 3\cos[2\pi \times 10^6 t + 2\sin(2\pi \times 10^3 t)]$. Determine its instantaneous frequency at time $t = 0.25$ ms.
 A) 6 MHz B) 6.2 MHz
 C) 62 MHz D) 1 MHz
- A signal with frequency f_m modulates a carrier f_c . Then the output of the AM-DSB-SC signal will contain frequencies
 A) $2f_c$ and $f_c \pm f_m$ B) f_c and $f_c \pm 2f_m$
 C) $f_c \pm f_m$ D) f_c and $f_c \pm f_m$
- A carrier is modulated to a depth of 40%. The percentage increase in the transmitted power is
 A) 40% B) 20%
 C) 16% D) 8%
- Which modulator is used to generate AM-DSB-SC
 A) Balanced modulator B) Square law modulator
 C) Armstrong phase modulator D) Envelope detector
- A transmitter puts out a total power of 3 watts at 30% modulation. How much power is contained in each of the sidebands ?
 A) 3 W B) 0.0645 W
 C) 0.09 W D) 2.87 W
- An amplitude modulated signal has a carrier frequency of 100 kHz. The upper sideband is transmitted at 106 kHz. The bandwidth required for the AM signal to transmit is _____
 A) 100 kHz B) 12 kHz
 C) 106 kHz D) 6 kHz
- _____ arises in electronic devices such as diodes and transistors because of the current flow in these devices.
 A) White noise B) Thermal noise
 C) Shot noise D) Pink noise

10. White noise of magnitude $0.001 \mu\text{W/Hz}$ is fed to an RC LPF with $R = 1000 \text{ ohm}$ and $C = 0.1 \mu\text{F}$. The output noise power will be
 A) $1 \mu\text{W}$ B) 1.3 mW
 C) 2.5 mW D) $2.5 \mu\text{W}$
11. Systems at temperature equilibrium will have average energy due to thermal agitation which is equal to _____ joules/degree of freedom.
 A) $kT/2$ B) kT/C
 C) $2kT$ D) kTB
12. If N -port networks are cascaded, then the overall noise figure is predominantly due to the
 A) first $N-1$ stages B) first two stages
 C) first stage D) $N/2$ stages
13. For transition between NBFM and WBFM, the value of β is chosen as
 A) >1 B) 0.75
 C) 0.5 D) 0.25
14. The power spectral density of the output noise _____ with frequency.
 A) increases rapidly B) decreases rapidly
 C) remains constant D) logarithmically decreases
15. The pre-emphasis and de-emphasis circuits are widely used in FM transmitter and receiver to effectively increase the _____.
 A) Bandwidth B) output SNR
 C) input SNR D) transmit power
16. While using coherent detection for DSB-SC signal, the LPF output will contain
 A) the message signal and inphase noise component B) the message signal and quadrature noise component
 C) only the message signal D) the message signal and both the inphase and quadrature noise components
17. The sampling rate of $x(t) = 2\cos(500\pi t)\cos(1000\pi t)$ is
 A) 750 Hz B) 1000 Hz
 C) 1500 Hz D) 500 Hz
18. A signal $x(t) = 8\sin(600t)$ is sampled and quantized using 10-bit PCM. The signal to noise ratio will be
 A) 60 dB B) 50 dB
 C) 68.2 dB D) 61.8 dB
19. A speech signal with a minimum frequency of 4 kHz and a maximum amplitude of 2 V is applied to a DM with its bit rate at 25 kbps . What should be the step size?
 A) $\leq 2 \text{ V}$ B) $\geq 0.5 \text{ V}$
 C) $\geq 2 \text{ V}$ D) $\leq 0.5 \text{ V}$
20. The mean square value of quantization noise in PCM is
 A) $S^2/12$ B) $S/2$
 C) $S^2/4$ D) $S/12$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) Draw the waveform of an amplitude modulated signal and derive expressions for (i) Modulation index (ii) Total power in the modulated wave (iii) Unmodulated carrier power (iv) Power in the sideband (v) Ratio of total power to carrier power.
- b) The carrier frequency $c(t) = 80\cos(2\pi \times 10^6 t)$ is frequency modulated by the signal $m(t) = 4\cos(2\pi \times 10^4 t)$. The frequency deviation is 20 kHz .
 (i) Determine the amplitude and frequency of all signal components that have a power level of at least 8% of the power of FM signal.
 (ii) What is the bandwidth of the FM signal?

Marks BT* CO* PO*

6 L*2 1 1

6 L3 2 1

2.	c)	Explain the use of envelope detector for the demodulation of AM wave.	4	L2	1	1
	a)	Consider a single tone modulation and derive an expression for the AM signal as well as its spectrum. Plot the spectrum and show the various components.	8	L2	1	1
	b)	A signal $m(t)$ frequency modulates a 100 kHz carrier to produce the following NBFM signal $x_{NBFM}(t) = 5 \cos(2\pi \times 10^5 t + 0.005 \sin(2\pi \times 10^4 t))$ Generate (block diagram & design) the WBFM signal with a carrier frequency of 120 MHz and a frequency deviation of 100 kHz. Assume that the following are available for the design: - Frequency multipliers of any (integer) value. - A local oscillator whose frequency can be tuned to any value between 50 MHz to 150 MHz. - An ideal BPF with tunable centre frequency and bandwidth. The block diagram must clearly specify the parameters at all logical points.	8	L3	2	1
3.	a)	Considering the characteristics of ideal slope circuit and using relevant mathematical expressions, show how the FM signal is demodulated using balanced frequency discriminator.	8	L2	2	1
	b)	A carrier wave of amplitude 5 V and frequency 100 MHz is frequency modulated by a sinusoidal signal of amplitude 5 V and frequency 20 kHz. The frequency sensitivity is 2 kHz/V. Sketch the spectrum of the frequency modulated signal.	4	L3	2	1
	c)	Discuss the effect of phase error in the local carrier in the synchronous detection of SSB-SC signal.	4	L2	1	1
		Unit – II				
4.	a)	Explain the concept of noise bandwidth. Show that the noise bandwidth of an RC filter is $\frac{\pi}{2}$ times its 3-dB bandwidth.	5	L2	3	1
	b)	Consider a receiver made up of the following three stages: The input stage is a pre-amplifier with a gain of 20 dB and a noise figure of 6 dB. The second stage is a 3-dB lossy network. The output stage is an amplifier with a gain of 60 dB and a noise figure of 16 dB. - Find the overall noise figure of the system - Repeat (i) with the pre-amplifier removed.	4	L3	3	1
5.	c)	Derive an expression for the figure of merit for AM DSB-SC signal using coherent detector.	7	L2	4	1
	a)	Discuss the effect of noise in SSB signal and derive an expression for its figure of merit.	8	L2	4	1
	b)	With proper mathematical analysis, show that the figure of merit for FM signal is $\frac{3}{2} \beta^2$.	8	L2	4	1
6.	a)	Explain the pre-emphasis and de-emphasis network with proper analysis and derive an expression for the improvement ratio.	8	L2	4	1
	b)	What is the need of receiver model? Explain the receiver model used to study the effect of noise in AM system and define the following (i) Input signal-to-noise ratio (ii) Output signal-to-noise ratio (iii) Channel signal-to-noise ratio (iv) Figure of merit.	6	L2	4	1
	c)	An FM signal $s(t) = 10 \cos[2\pi \times 10^8 t + 8 \sin(2\pi \times 10^3 t)]$ is received at the receiver. Calculate the figure of merit of this receiver	2	L3	4	1

Unit – III

7. a) Consider the analog signal $x(t) = 3 \cos(100 \pi t)$
- Determine the minimum sampling rate required to avoid aliasing.
 - Suppose that the signal is sampled at the rate of 200 Hz, what is the discrete time signal obtained after sampling?
 - Suppose that the signal is sampled at the rate of 75 Hz, what is the discrete time signal obtained after sampling?
 - What is the frequency of a sinusoid that yields samples identical to those obtained in part (iii) ?
- b) With proper assumptions, derive an expression for the signal-to-noise ratio of a PCM system.
- c) Consider a test signal $m(t) = A \tanh(\beta t)$ defined by a hyperbolic tangent function: where A and β are constants. Determine the minimum step size for delta modulation of this signal, which is required to avoid slope overload.
8. a) A signal $x(t) = \cos(6\pi \times 10^3 t)$ is sampled at a rate of 10 kHz. The sampled signal is then applied to an ideal LP reconstruction filter. Find an expression for the output and plot the spectrum.
- b) For a sinusoidal modulating signal $x(t) = A_0 \cos(2\pi f_0 t)$ show that output signal-to-quantization noise ratio in a Delta modulation system under no slope overload is given by
- $$[SNR]_0 = \frac{3f_s^3}{8\pi^2 f_0^2 f_M}$$
- c) Determine the Nyquist sampling rate for the signal
- $$m(t) = \frac{\sin(300\pi t)}{\pi t} \times \frac{\sin(500\pi t)}{\pi t}$$

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
